

Understanding Retrieval-Augmented Generation (RAG)

Retrieval-Augmented Generation, or RAG, is a technique that combines a retrieval system with a language model to produce more accurate and grounded answers. Instead of relying purely on what the model memorized during training, RAG first retrieves relevant documents from a knowledge base and then feeds them to the language model as context.

A typical RAG pipeline has three main stages: ingestion, retrieval, and generation. In the ingestion stage, documents are loaded, parsed, and split into smaller chunks. Each chunk is converted into a vector embedding and stored in a vector database. In the retrieval stage, a user query is embedded the same way, and the most similar chunks are fetched from the vector store.

Finally, in the generation stage, the retrieved chunks are passed to a large language model along with the original question, and the model generates an answer grounded in that retrieved context. This reduces hallucination compared to asking the model a question with no supporting evidence.

Production RAG systems add more layers on top of this basic flow, such as reranking retrieved chunks for relevance, monitoring latency and cost at each stage, handling failures gracefully with fallback responses, and evaluating answer quality against a golden dataset of known correct answers.